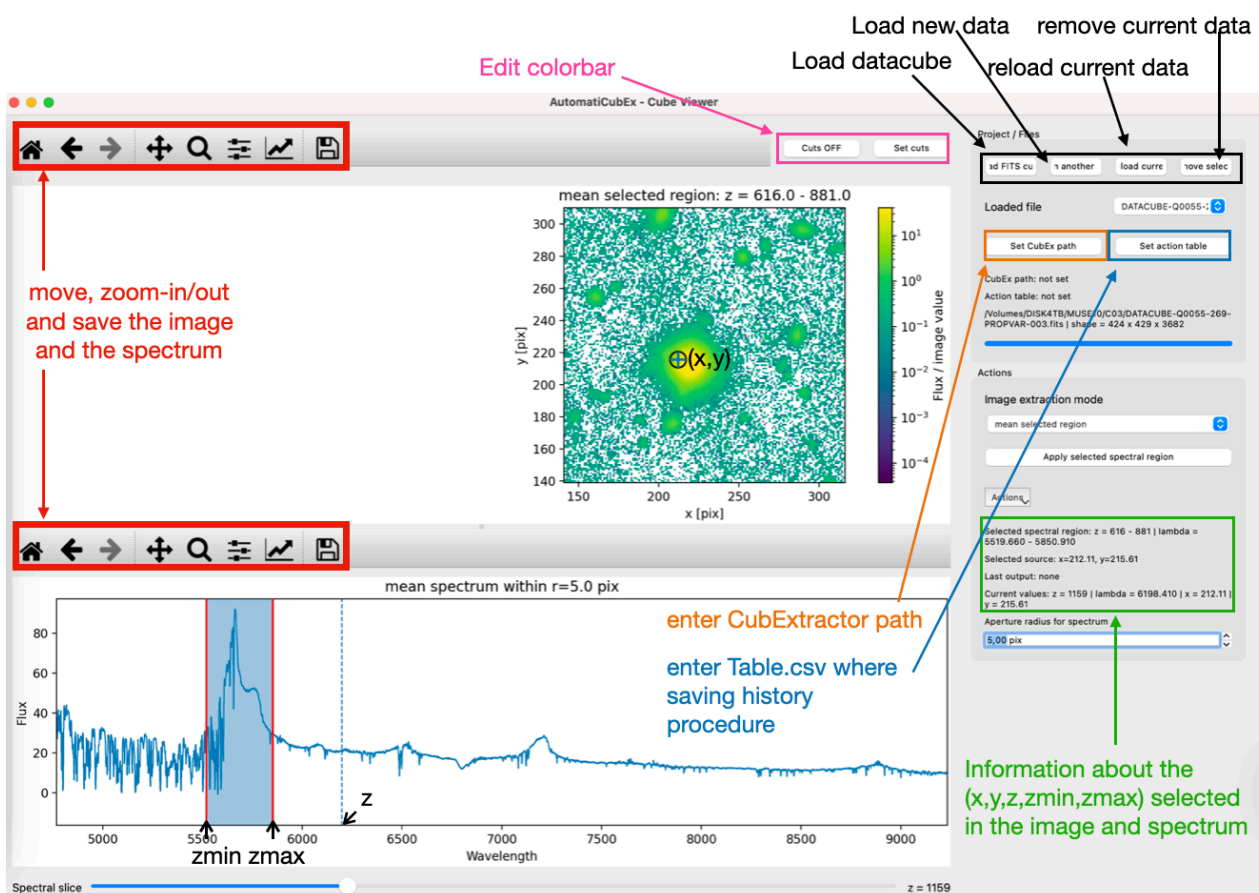


AutomatiCubEx: A Visual Step-by-Step Guide for Extended Emission Extraction

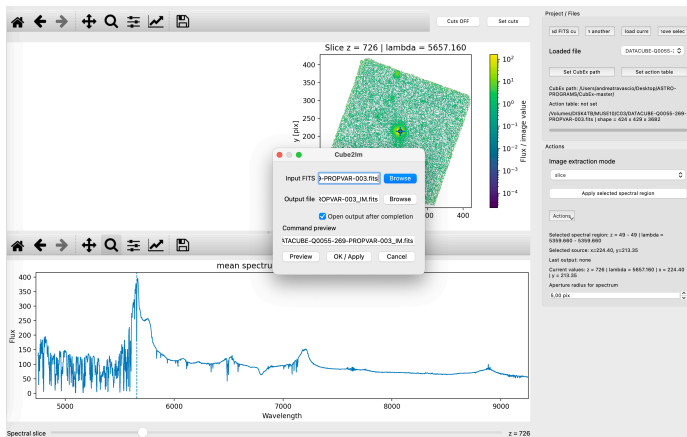
This guide illustrates a standard workflow to extract faint extended emission around luminous sources such as quasars and galaxies using AutomatiCubEx. The software combines an interactive GUI with CubEx/CubExtractor tools, allowing users to visualize datacubes, inspect spectra, apply processing routines, and generate scientific products such as masks and moment maps.

1. Understanding the GUI Interface

The main AutomatiCubEx interface provides simultaneous visualization of the image and spectrum. Users can zoom interactively, inspect specific pixels or apertures, modify image cuts similarly to ds9/QFitsView, and load multiple FITS files. The upper panel displays the spatial image, while the lower panel shows the extracted spectrum. Spectral selections immediately update the image display, allowing users to isolate faint extended emission regions.

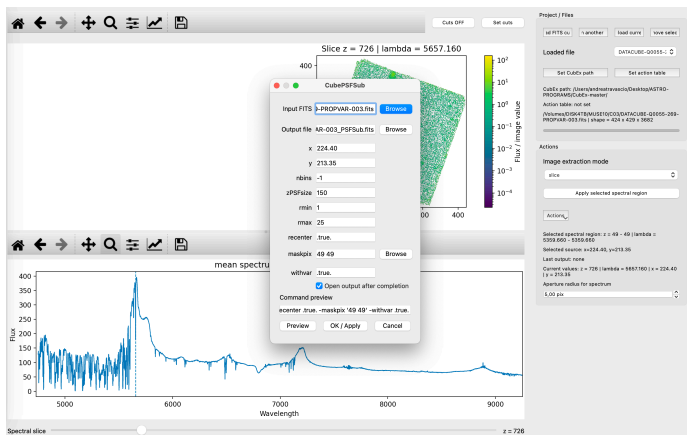


2. Generating a White-Light Image with Cube2Im



The first processing step is usually the generation of a white-light image (WLI) from the original datacube using Cube2Im. This image is useful for source identification and mask generation. Only the input cube and output image are required for this step.

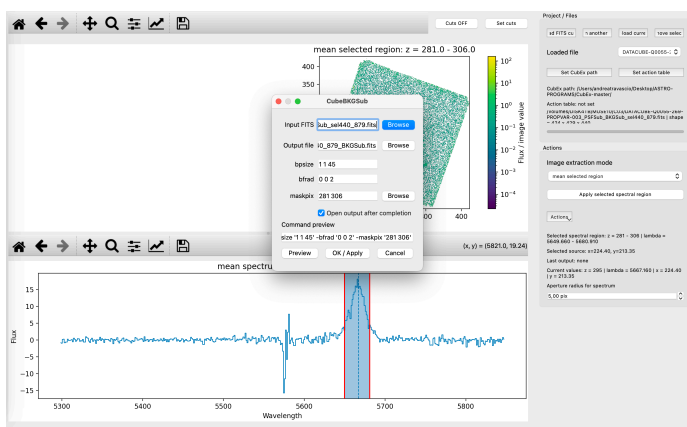
3. PSF Subtraction Workflow



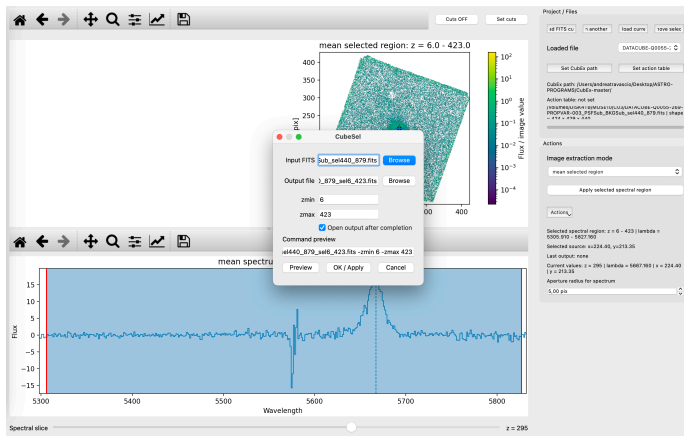
A standard analysis generally requires two PSF subtraction runs. First, the subtraction is performed without spectral masking in order to identify the approximate wavelength range of the extended emission. Afterwards, the same procedure is repeated while masking the spectral region containing the nebular emission. In AutomatiCubEx, this masking region can be selected directly from

the spectrum panel. The QSO centroid should be carefully selected before running the PSF subtraction. Parameters such as PSFsize and rmax strongly affect the quality of the empirical PSF model.

4. Background Subtraction and CubeSel

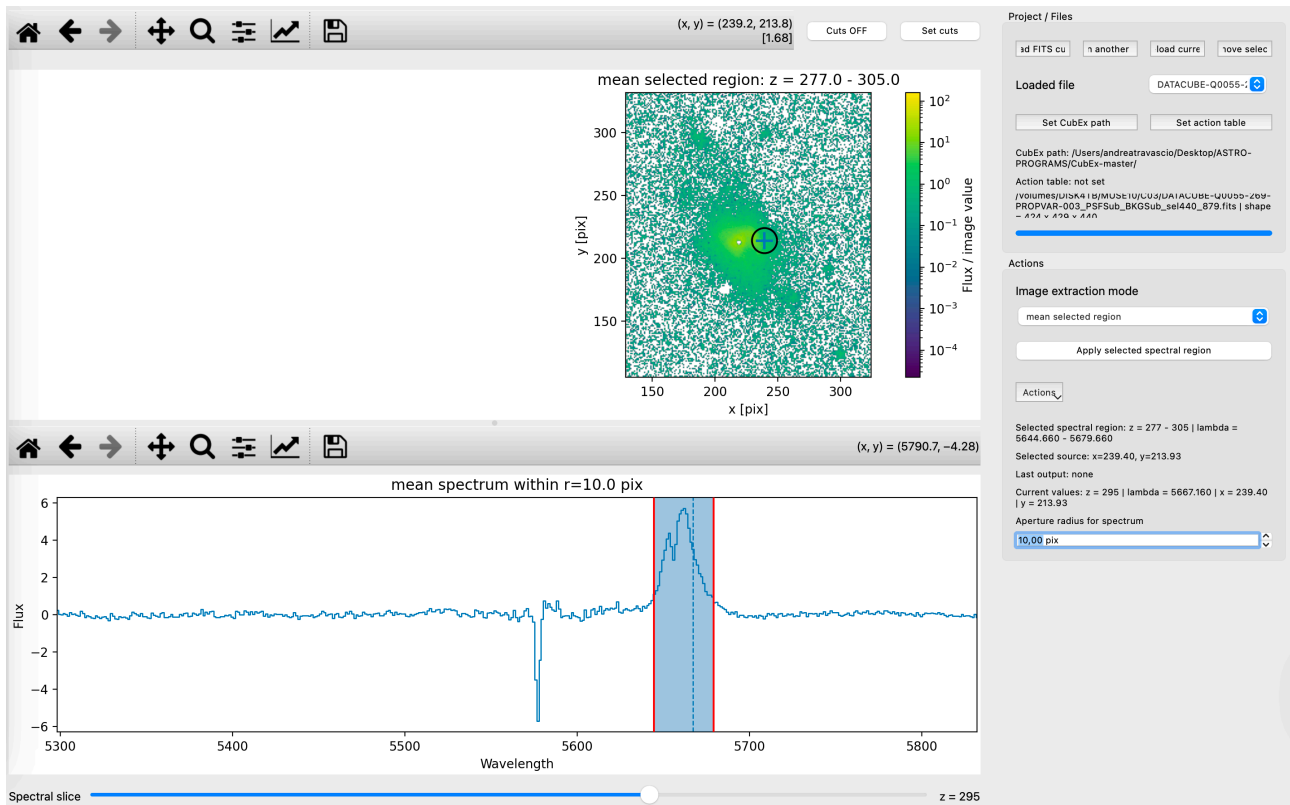


After PSF subtraction, the background subtraction is applied. The same spectral masking region should normally be used in order to avoid biasing the background estimate. CubeSel is then used to reduce the spectral size of the datacube, keeping only the wavelength region containing the emission line and a small continuum region around it.



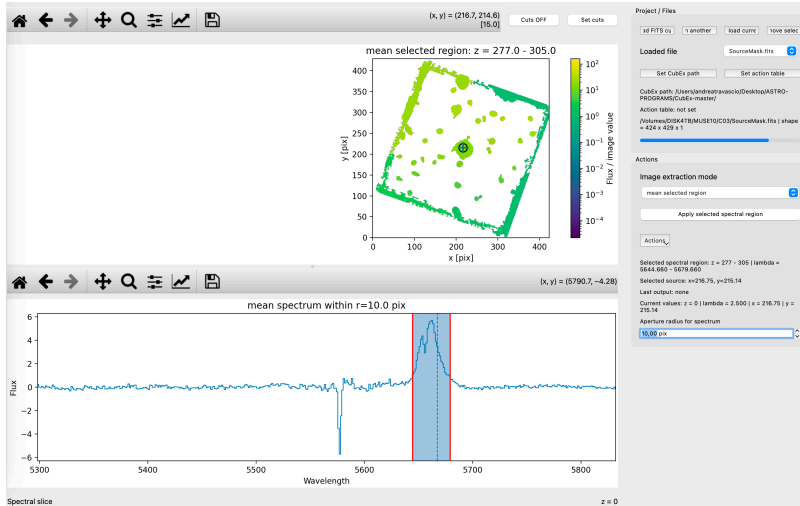
5. Interactive Exploration of the Emission

The GUI allows the user to select a spatial aperture and inspect the corresponding spectrum interactively. By selecting a spectral interval, the user can generate a mean or median image of the selected wavelength range. This is extremely useful for locating faint extended emission and determining the proper spectral range for extraction.

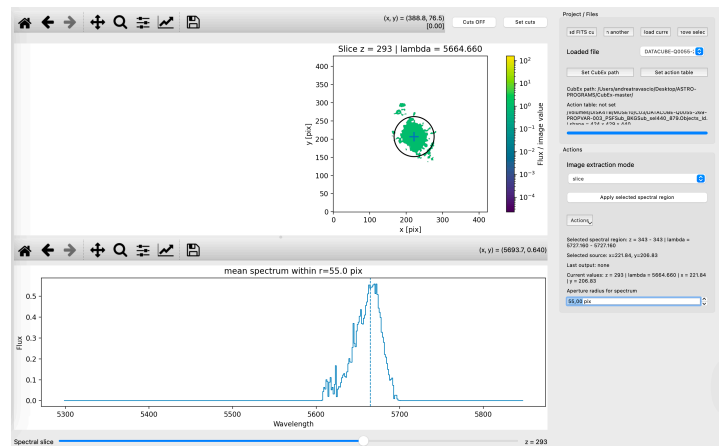
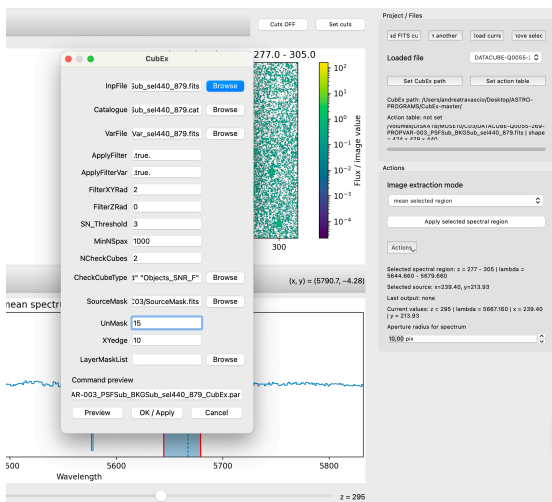


6. CreateMask and CubEx Extraction

CreateMask generates a source mask from the white-light image. Each source is assigned an ID number. The user should identify the source around which the extended emission is searched and remember its ID.



CubEx is then applied to the PSF-subtracted, background-subtracted, and CubeSel-cut datacube. Important parameters include: - SN_Threshold - MinNVox / MinNSpax - SourceMask - UnMask source ID CubEx produces a 3D segmentation mask identifying extended structures (right panel).



7. Binarize3DMask and CreateMaps

The Binarize3DMask tool allows the user to isolate one or multiple detected regions by converting selected IDs to 1 and all remaining voxels to 0. Finally, CreateMaps generates: - Surface brightness maps - Velocity maps - Kinematic products (right panel) using the final binary 3D mask.

